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Experiment No. 10

Aim: Explore the Cryptocurrency Landscape

Theory:

Cryptocurrency Landscape

1. Definition – Cryptocurrencies are decentralized digital assets built on blockchain networks, enabling secure, transparent, and immutable transactions without central authority. 2. Market Diversity – Thousands of cryptocurrencies exist, each serving unique roles: ○ Store of Value – Bitcoin, Litecoin.
    - Utility Tokens – Chainlink, Ethereum.
    - Stablecoins – USDT, USDC.
    - Governance Tokens – UNI, AAVE.
3. Volatility – Prices are highly sensitive to market sentiment, technological developments, and regulatory updates.
4. Data Analysis Importance – Tracking historical trends, market capitalization, and trading volume is essential for understanding asset performance.
5. Advancements – Growing use of DeFi platforms, NFT ecosystems, layer-2 scaling, and cross-chain protocols.

Data Collection Process

The program used the CoinGecko API to automate historical data fetching for Chainlink. The retrieved data was processed with Pandas to compute:

- Daily Returns – Percentage change between consecutive days to measure volatility.
- Moving Averages (MA7, MA30) – Short-term and long-term trend indicators. ● Cumulative Returns – Growth factor over time.
- Rolling Volatility – Short-term risk measurement based on daily return fluctuations.

Visualization & Dashboard

Using Matplotlib, a six-panel dashboard was created to display:

1. Price Trend – Daily price movement over the selected period.
2. Price vs Volume – Correlation between trading activity and price changes.
3. Daily Returns Distribution – Statistical spread of daily gains/losses.
4. Moving Averages – Technical analysis trend lines for price movement.
5. Cumulative Returns – Overall growth performance.
6. Rolling Volatility – Market risk trends over time.

Technological Analysis

Beyond market data, the study covered advancements in blockchain technology, including: ● Consensus Mechanisms – PoW (Proof of Work), PoS (Proof of Stake), and hybrid systems. ● Scalability Solutions – Layer-2 protocols, sharding, and sidechains to increase throughput. ● Interoperability Protocols – Chainlink's CCIP, Polkadot, and Cosmos enabling cross-chain data exchange.

- Privacy Features – Zero-knowledge proofs (zk-SNARKs) and ring signatures to ensure transaction

confidentiality.

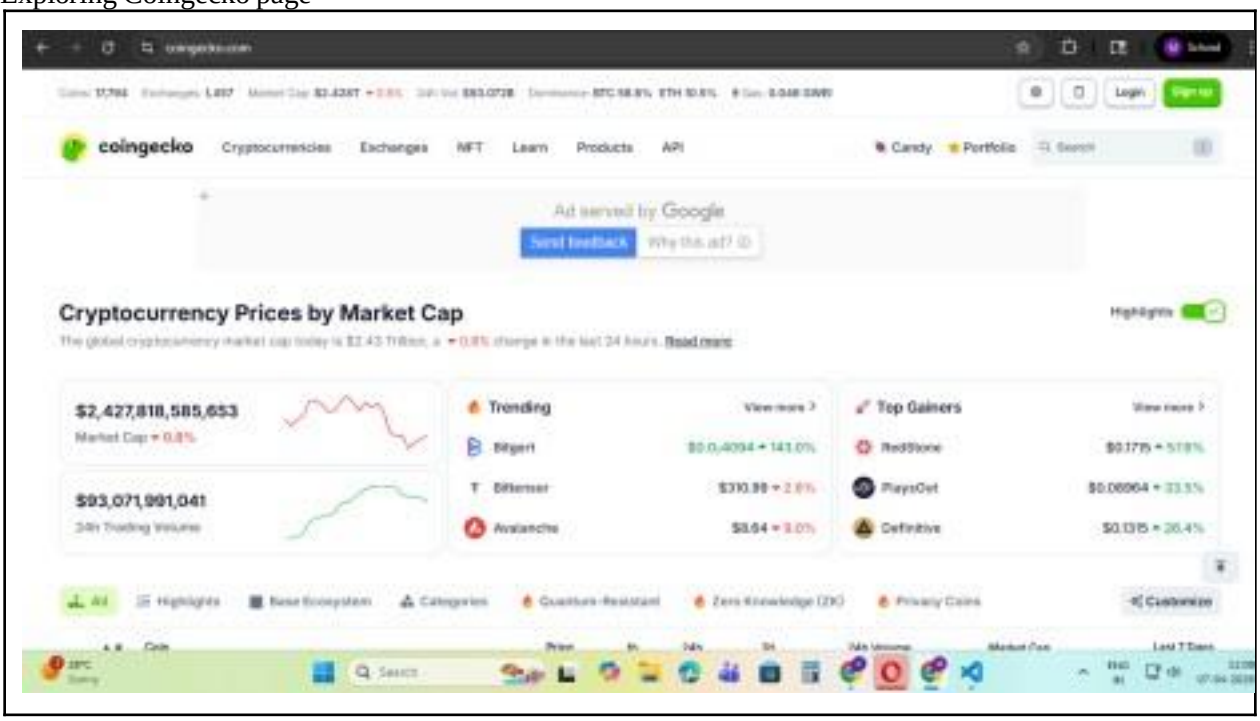
Chainlink's roadmap highlights staking implementation, advanced oracle features, and enhanced cross-chain capabilities, aligning with industry trends toward decentralization, interoperability, and security.

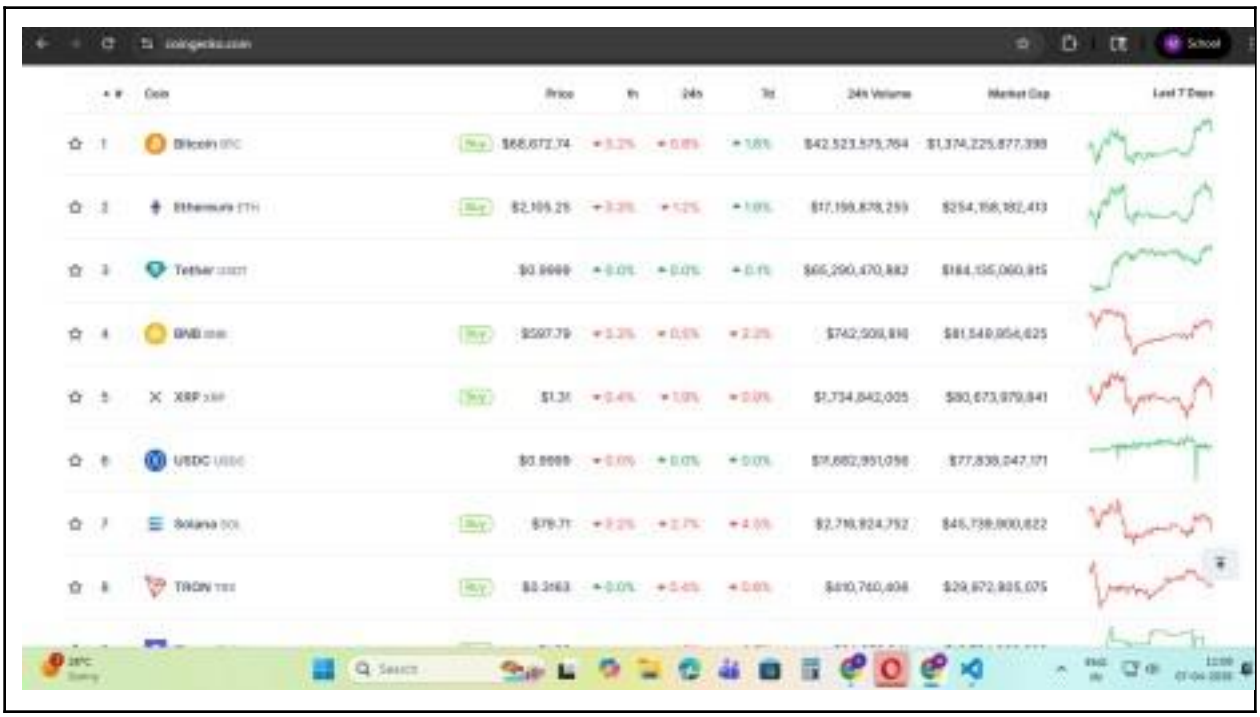
Market Trends & Adoption

- Growing Merchant Acceptance – More businesses accepting cryptocurrency payments.
- Wallet Growth – Increasing downloads and installations of crypto wallets.
- Network Activity – Higher transaction volumes and active addresses, signaling strong adoption.
- Integration with Traditional Finance – Partnerships between crypto platforms and fintech/payment providers.
- Regulatory Developments – Countries exploring frameworks for safe cryptocurrency adoption.

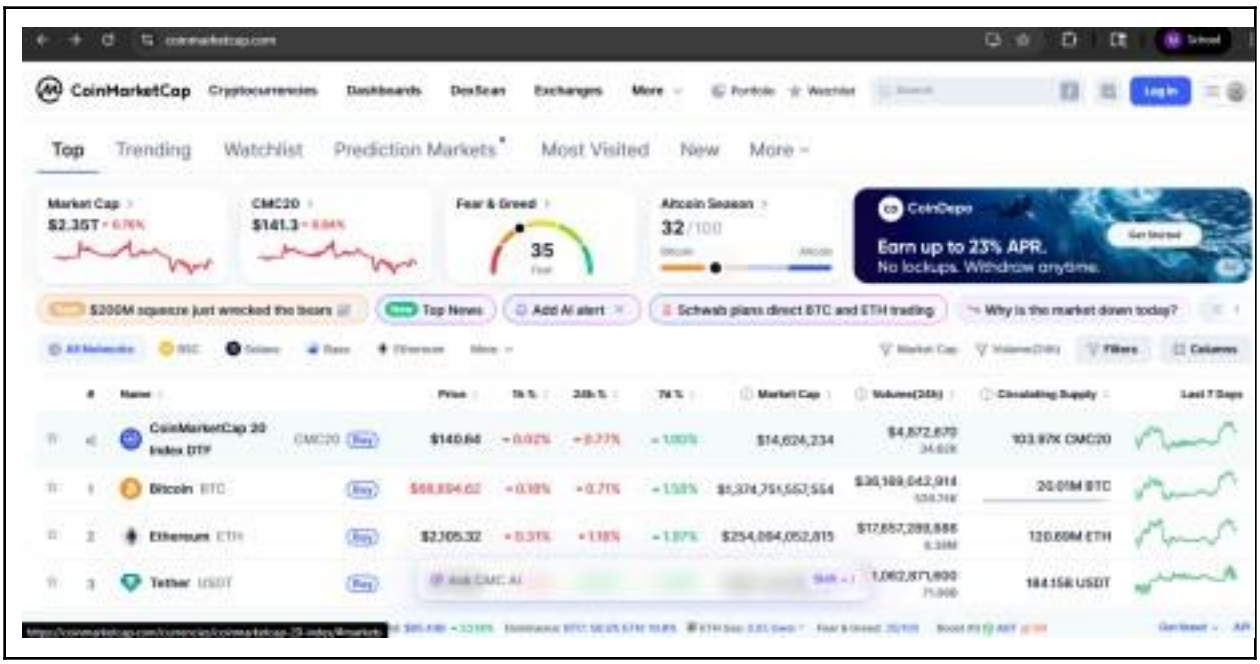
Implementation:

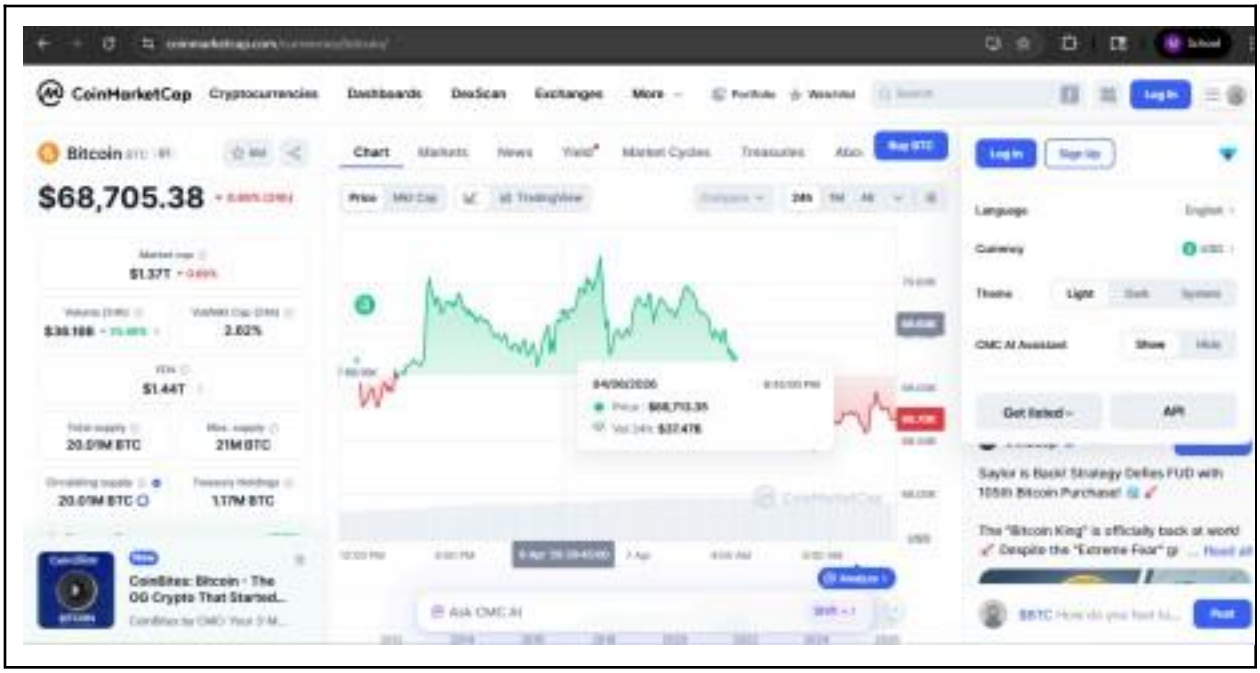
Exploring Coingecko page-





Exploring CoinMarketCap





Code:

(Imported APIs to extract data of market)

```
import requests
import matplotlib.pyplot as plt
import pandas as pd

# ------------ Fetch Data ------------
def get_historical_data(coin_id, days='180', currency='usd'):
    url = f'https://api.coingecko.com/api/v3/coins/{coin_id}/market_chart'
    params = {'vs_currency': currency, 'days': days}
```

```
    response = requests.get(url, params=params)
    return response.json()

# ---------- Dashboard Function ----------
def chainlink_dashboard(days='180'):
    coin_id = 'bitcoin'
    data = get_historical_data(coin_id, days)

    # Create DataFrame
    prices_df = pd.DataFrame(data['prices'], columns=['timestamp', 'price'])
    volumes_df = pd.DataFrame(data['total_volumes'], columns=['timestamp', 'volume'])

    prices_df['timestamp'] = pd.to_datetime(prices_df['timestamp'], unit='ms')
    volumes_df['timestamp'] = pd.to_datetime(volumes_df['timestamp'], unit='ms')

    df = pd.merge(prices_df, volumes_df, on='timestamp')
    df.set_index('timestamp', inplace=True)

    # Calculate extra metrics
    df['returns'] = df['price'].pct_change()
    df['MA7'] = df['price'].rolling(window=7).mean()
    df['MA30'] = df['price'].rolling(window=30).mean()
    df['cumulative'] = (1 + df['returns']).cumprod()
    df['volatility'] = df['returns'].rolling(window=7).std() * 100

    # ---------- Plot Dashboard ----------
    fig, axes = plt.subplots(3, 2, figsize=(16, 12))
    fig.suptitle(f'{coin_id.capitalize()} Dashboard (Last {days} Days)', fontsize=16, fontweight='bold')

    # 1. Price Trend
    axes[0, 0].plot(df.index, df['price'], color='blue')
    axes[0, 0].set_title('Price Trend')
    axes[0, 0].set_ylabel('Price (USD)')
    axes[0, 0].set_xlabel('Year-Month') # Add xlabel
    axes[0, 0].grid(True)

    # 2. Price vs Volume
    axes[0, 1].plot(df.index, df['price'], color='blue', label='Price')
    axes[0, 1].bar(df.index, df['volume'], color='orange', alpha=0.3, label='Volume')
    axes[0, 1].set_title('Price vs Volume')
    axes[0, 1].set_xlabel('Year-Month') # Add xlabel
    axes[0, 1].set_ylabel('Price (USD) / Volume') # Add ylabel
    axes[0, 1].legend()
    axes[0, 1].grid(True)

    # 3. Daily Returns Distribution
    axes[1, 0].hist(df['returns'].dropna(), bins=50, alpha=0.7, color='purple')
    axes[1, 0].set_title('Daily Returns Distribution')
```

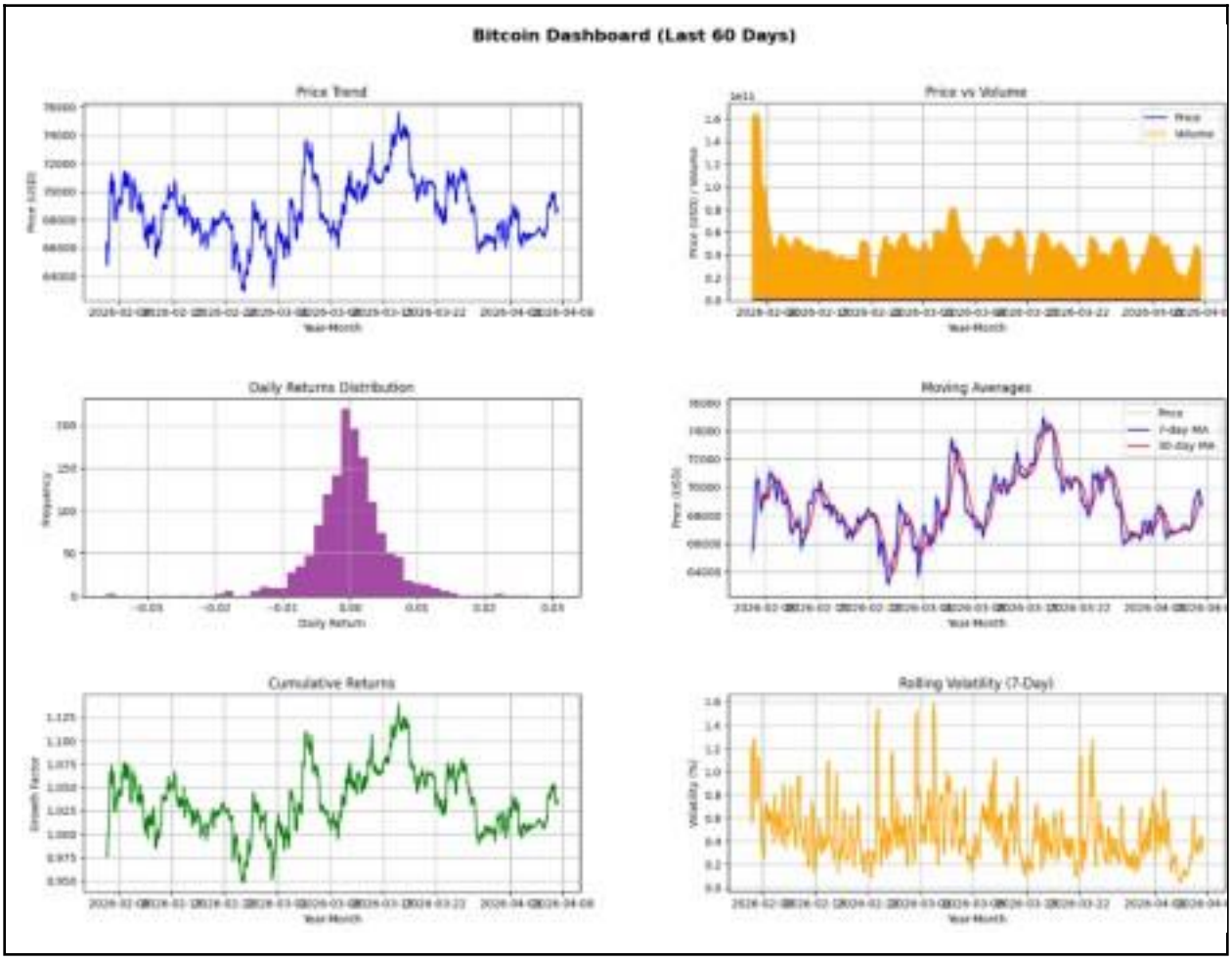
```
    axes[1, 0].set_xlabel('Daily Return')
    axes[1, 0].set_ylabel('Frequency')
    axes[1, 0].grid(True)

    # 4. Moving Averages
    axes[1, 1].plot(df.index, df['price'], color='lightgray', label='Price')
    axes[1, 1].plot(df.index, df['MA7'], color='blue', label='7-day MA')
    axes[1, 1].plot(df.index, df['MA30'], color='red', label='30-day MA')
    axes[1, 1].set_title('Moving Averages')
    axes[1, 1].set_xlabel('Year-Month')
    axes[1, 1].set_ylabel('Price (USD)')
    axes[1, 1].legend()
    axes[1, 1].grid(True)

    # 5. Cumulative Returns
    axes[2, 0].plot(df.index, df['cumulative'], color='green')
    axes[2, 0].set_title('Cumulative Returns')
    axes[2, 0].set_ylabel('Growth Factor')
    axes[2, 0].set_xlabel('Year-Month')
    axes[2, 0].grid(True)

    # 6. Volatility
    axes[2, 1].plot(df.index, df['volatility'], color='orange')
    axes[2, 1].set_title('Rolling Volatility (7-Day)')
    axes[2, 1].set_ylabel('Volatility (%)')
    axes[2, 1].set_xlabel('Year-Month')
    axes[2, 1].grid(True)

    plt.tight_layout(rect=[0, 0, 1, 0.96])
    plt.subplots_adjust(hspace=0.5, wspace=0.3) # Adjust vertical and horizontal spacing
    plt.show()
# ---------- Run Dashboard ----------
chainlink_dashboard('60')
```



Conclusion:

By combining automated market data processing, visual trend analysis, blockchain technology evaluation, and adoption metrics, this experiment presents a holistic view of the cryptocurrency ecosystem. It demonstrates how blockchain's technical foundations connect with real-world market behavior, providing insights into both current trends and future advancements in the digital asset industry.